Xuanjun Chen

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Summary

Xuanjun is a Ph.D. candidate at National Taiwan University, advised by Profs. Hung-yi Lee and Jyh-Shing Roger Jang. He focuses on **Speech & Audio Processing** and **Large Language Models**, specifically audio understanding, audio generation, agentic LLMs, and deepfakes. He has authored **15+ first/co-first papers** in Speech/NLP venues (e.g., COLM, TIST, TASLP, ICASSP, INTERSPEECH, ASRU, SLT), and is a core contributor to DeSTA2.5-Audio (TASLP) and SUPERB benchmarks (ICLR, ACL Findings). He holds an Industry PhD Fellowship, NVIDIA Academic Grant, and IEEE SPS Scholarship, alongside an IEEE ASRU 2025 Best Student Paper Nomination and Best/Outstanding Paper awards at ROCLING 2025 and ISCSLP 2026.

Education

National Taiwan University, Ph.D. in Communication Engineering, GPA: 4.3/4.3 — Sep 2023 – Present
National Taiwan University, M.S. in Computer Science & Information Engineering (CSIE), GPA: 4.19/4.3 — Jan 2023
National Taiwan University of Science and Technology, B.S. in CSIE, 3-time Dean's List, GPA: 4.11/4.3 — Jun 2020

Research Experience

NVIDIA Academic Grant Program, Student Researcher, Taipei Taiwan — Oct 2026 – Mar 2027
- **Audio Generation.** Leading research on fine-grained text-to-audio model to enhance audio generation capabilities.

Shanda Group, Research Intern (Remote), Tokyo Japan — Apr 2026 – Jul 2026
- **Speech Generation.** Investigated whether LLM agents can automate reinforcement learning for Text-to-Speech models. [1]

National Taiwan University, Research Assistant (Ph.D. & M.S.), Taipei Taiwan — Oct 2020 – Present
- Published over 20 peer-reviewed publications with **15 as first/co-first author**, alongside 5 more currently under review.
- **Speech and Audio Language Models.** Analyzed the post-training dynamics of large audio language models [2], authored a high-impact survey [7], and served as a core contributor to DeSTA2.5-Audio [3], Dynamic-SUPERB [4], and Codec-SUPERB [5] [6].
- **Retrieval and Agentic LLMs.** Pioneered a dual-loop agentic framework for optimizing benchmark generation [8]; improved retrieval via graph navigation [9] and delta planning [10], alongside a systematic RAG study for historical archives [11].
- **Deepfake Speech.** Conducting research on speech anti-spoofing, building upon prior work on audio deepfake detection across various scenarios [12]–[21] and audio-visual learning [22]–[24], including a system ranked 3rd of 42 teams in ASVspoof 2021 (LA track) [25].

Research Highlights

*Total citations **600+** as of Oct 2026.*

- MetaBench-Harness: Unlocking End-to-End Optimization of Benchmark Harnesses — *ICLR '27 (Submitted)*
  TL;DR: *A dual-loop framework where a meta-harness refines the harnesses that evolve benchmarks hard enough for frontier LLMs.*
- Can LLM Agents Automate Reinforcement Learning for Text-to-Speech? — *ICASSP '27 (Submitted)*
  TL;DR: *An LLM agent reconstructs and improves an RL post-training recipe for zero-shot TTS, cutting failure cases by half.*
- What Does LALM Post-training Learn? Rethinking Audio Understanding and Reasoning Capabilities — *NAACL '27 (Submitted)*
  TL;DR: *Much of the post-training gain comes from fitting answer formats and textual cues, not from better audio understanding.*
- Only Ask What You Don't Know: Grounded Delta Planning for Efficient Multi-step RAG — *COLM '26*
  TL;DR: *Plans only for missing information after a first retrieval; top multi-hop QA accuracy at 22–68% lower cost.*
- CodaRAG: Connecting the Dots with Associativity Inspired by Complementary Learning — *ACM TIST '26*
  TL;DR: *Brain-inspired RAG that chains dispersed evidence via associative graph navigation; +7–10% recall on GraphRAG-Bench.*
- Towards Generalized Source Tracing for Codec-Based Deepfake Speech — *ASRU '25* · Best Student Paper Nominee (Top 2.5%)
  TL;DR: *SASTNet fuses Whisper semantics with Wav2vec2/AudioMAE acoustics to trace CodecFake speech back to its source codec.*

Open Source Contributions

Lead
- **CodecFake+** Largest deepfake dataset from neural audio codec-based TTS (31 codecs); 4.3K+ downloads [HF] — *TASLP '26*
- **DFADD** Mentored students on the first deepfake dataset from diffusion/flow-matching TTS; 9.1K+ downloads [HF] — *SLT '24*
- **SingGraph** Achieved SOTA on SingFake detection, fusing MERT and wav2vec 2.0; 24 GitHub stars [GitHub] — *INTERSPEECH '24*
- **MTDVocaLiST** Compact audio-visual lip-sync model distilled from VocaLiST; 29 GitHub stars [GitHub] — *ICASSP '24*

Collab.
- **DeSTA2.5-Audio** (Core Contributor) 142 GitHub stars, HF model (28.7K+) and data (11.0K+ downloads) [GitHub] — *TASLP '26*
  – *Curated training data for audio deepfake understanding as part of an audio LLM project in collaboration with NVIDIA*
- **Dynamic-SUPERB Phase-2** (Task Editor) 201 GitHub stars, 325 HF datasets (289K+ downloads) [GitHub] — *ICLR '25*
  – *Reviewed and standardized community-submitted tasks for the 180-task audio understanding benchmark*
- **Codec-SUPERB** (Core Contributor) 310 GitHub stars and a public leaderboard [Website] — *ACL Findings '24 & SLT '24*
  – *Built task- and signal-level metrics to benchmark several codecs on the neural audio codec leaderboard*

Selected Honors, Service & Skills

Honors	**Industry PhD Fellowship** Awarded via competitive global selection by an Industry sponsor	2026
	NVIDIA Academic Grant GPU compute resources via competitive selection by NVIDIA	2026
	IEEE SPS Scholarship One of dozens of recipients worldwide from the IEEE Signal Processing Society	2026
	Google Student Travel Grant Sponsored by Google Asia Pacific to present at ICASSP 2024	2024
	CTCI Foundation Sci. & Tech. Scholarship Bursary Award 2024 & Research Scholarship 2025	2024–2025
	Dean's List Certificate of Achievement for the top 5% of CSIE students, Taiwan Tech	Fall '18, Fall '19, Spr. '20
Service	**Co-Organizer,** Responsible Speech and Audio Generative AI, Special Session at IEEE ASRU 2025	2025
	Technical Committee, Codec Speech Proc. Universal PERformance Benchmark Challenge, IEEE SLT 2024	2024
	Reviewer: AAAI, ICLR, NeurIPS, ICML; ACL, EMNLP, NAACL, COLM; ICASSP, INTERSPEECH, ASRU, SLT, etc.	2023–Now
Skills	**Coding & Toolkit:** Python, LaTeX; PyTorch, PyTorch Lightning, Hugging Face Transformers	

Selected Publications

* *equal contribution;* † *co-second author;* ‡ *contributing author*

Speech & Audio Language Models and Generation

[1] **X. Chen**, et al., "Can LLM Agents Automate Reinforcement Learning for Text-to-Speech?" *ICASSP 2027 (Submitted)*

[2] H.H. Chen*, **X. Chen***, et al., "What Does LALM Post-training Learn? Rethinking Audio Understanding and Reasoning Capabilities," *NAACL 2027 (Submitted)*

[3] K.H. Lu, Z. Chen, ... **X. Chen**‡, ... H. Lee, "DeSTA2.5-Audio: Toward General-Purpose Large Audio Language Model with Self-Generated Cross-Modal Alignment," *TASLP, 2026*

[4] C.Y. Huang, ... **X. Chen**‡, ... S. Watanabe, H. Lee, "Dynamic-SUPERB Phase-2: A Collaboratively Expanding Benchmark for Measuring the Capabilities of Spoken Language Models with 180 Tasks," *ICLR 2025*

[5] H. Wu*, ... **X. Chen**†, ... H. Lee, "Codec-SUPERB: An In-Depth Analysis of Sound Codec Models," *Findings of ACL 2024*

[6] H. Wu, **X. Chen**, et al., "Codec-SUPERB @ SLT 2024: A Lightweight Benchmark for Neural Codec Models," *SLT 2024*

[7] H. Wu, **X. Chen**, et al., "Towards Audio Language Modeling – An Overview," *Technical Report, Feb. 2024*

Retrieval and Agentic Large Language Models

[8] **X. Chen**, et al., "MetaBench-Harness: Unlocking End-to-End Optimization of Benchmark Harnesses," *ICLR '27 (Submitted)*

[9] C.Y. Li*, **X. Chen***, et al., "CodaRAG: Connecting the Dots with Associativity Inspired by Complementary Learning," *ACM TIST, 2026*

[10] W.C. Chou*, **X. Chen***, et al., "Only Ask What You Don't Know: Grounded Delta Planning for Efficient Multi-step RAG," *COLM 2026*

[11] C. Lin*, B.H. Feng*, **X. Chen***, et al., "A Preliminary Study of RAG for Taiwanese Historical Archives," *ROCLING 2025* · Best Paper Award

Audio Deepfakes: Neural Codec, Singing Voice, Adversarial Attack, Conventional TTS/VC, and Audio-Visual Scenarios

[12] **X. Chen**, et al., "Why Is Spectrogram ResNet Still Effective in Singing Voice Deepfake Detection?" *ICASSP '27 (Submitted)*

[13] Y.A. Hsieh*, **X. Chen***, et al., "On Bona Fide Audio Discrepancies in Singing Voice Deepfake Detection," *ICASSP '27 (Submitted)*

[14] **X. Chen***, et al., "CodecFake+: Codec-Based Resynthesized Data as a Proxy for Detecting CodecFake Speech," *TASLP, 2026*

[15] **X. Chen**, et al., "Mitigating Proxy-to-Wild Domain Gap in Deepfake Speech," *ISCSLP 2026* · Outstanding Paper

[16] **X. Chen***, C.Y. Hu*, et al., "Joint Fullband-Subband Modeling for High-Resolution SingFake Detection," *INTERSPEECH 2026* · Long Oral

[17] **X. Chen**, et al., "How Does Instrumental Music Help SingFake Detection?" *ICASSP 2026*

[18] **X. Chen***, I.M. Lin*, et al., "Towards Generalized Source Tracing for Codec-Based Deepfake Speech," *ASRU 2025* · Best Student Paper Nominee

[19] **X. Chen**, et al., "Codec-Based Deepfake Source Tracing via Neural Audio Codec Taxonomy," *INTERSPEECH 2025*

[20] **X. Chen**, et al., "Singing Voice Graph Modeling for SingFake Detection," *INTERSPEECH 2024* · Oral

[21] **X. Chen**, et al., "Neural Codec-based Adversarial Sample Detection for Speaker Verification," *INTERSPEECH 2024*

[22] **X. Chen**, et al., "Localizing Audio-Visual Deepfakes via Hierarchical Boundary Modeling," *ISCSLP 2026*

[23] **X. Chen**, et al., "Multimodal Transformer Distillation for Audio-Visual Synchronization," *ICASSP 2024*

[24] **X. Chen**, et al., "Push-Pull: Characterizing the Adversarial Robustness for Audio-Visual Active Speaker Detection," *SLT 2022*

[25] Y.L. Liao*, **X. Chen***, et al., "Adversarial Speaker Distillation for Countermeasure Model on Automatic Speaker Verification," *SPSC 2022*